On the Nature of Subharmonics of the Electron Emission from Ultracold Plasmas

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One of the most interesting phenomena in the ultracold plasmas are multiple subharmonics of the electron emission observed after its irradiation by the monochromatic radiowaves. Unfortunately, the early interpretation of this phenomenon as the so-called Tonks–Dattner resonances (*i.e.*, actually the standing Langmuir waves) encountered a number of serious obstacles, such as a lack of the adequate boundary conditions, an incorrect dependence on the electron temperature, and an insensitivity to the shape of the cloud. Here, we suggest an alternative interpretation based on the quasi-classical multiphoton ionization of the 'secondary' Rydberg atoms formed in the expanding and cooling plasma clouds. As follows from our numerical simulations, the efficiency of such ionization exhibits a series of well-expressed peaks. Moreover, this process is evidently irrelevant to the boundary conditions and global shape of the cloud. Therefore, this should be a viable alternative to the earlier idea of Tonks–Dattner resonances.

I. INTRODUCTION

The physics of ultracold plasmas (UCP) is a relatively new branch of plasma physics, originated in the very late 1990's and early 2000's due to advances in the technology of laser cooling of gases and their capture in the magneto-optical traps[1–5]. They are the quasi-neutral systems of charged particles with electronic temperature from a few to several hundreds of Kelvin and the typical ionic temperature about 1 K or below. It is interesting to mention that theoretical studies of such plasmas were started even before their experimental realization became feasible; *e.g.*, review[6]. A new impact to the research of UCP was given by the subsequent experiments on their creation in the continuous regime, both in the supersonic gas-dynamic jets[7] and—most recently—by employing the standard technique of laser cooling[8].

The ultracold plasmas were considered initially, first of all, as a promising tool to achieve considerable values of the Coulomb coupling parameter (*i.e.*, a ratio of the potential and kinetic energies)[9]. However, a number of other interesting and unexpected phenomena were found in the course of their experimental studies. One of them are multiple subharmonics of the electron emission observed after irradiation of UCP by radiowaves. In other words, when the ultracold plasma bunch expands monotonically and interacts with a monochromatic electromagnetic wave, the flux of escaping electrons exhibits a substantially nonmonotonic behavior, with a series of well-expressed peaks.

In fact, the basic harmonic of this signal was found already in the first experiments with UCP and became the crucial evidence that a collective behavior of the electrons was really achieved[10]. A few years later, a whole series of the additional peaks was found in the more elaborated experiment[11]; and their interpretation became challenging. The first and most evident idea was that they were the subharmonics of the standing Langmuir waves excited in the plasma cloud of a finite size, or the so-called Tonks–Dattner (TD) resonances[12,13]. Namely, it was assumed that the cloud in the course of its expansion passed through a series of 'preferable' radii, which were favorable for excitation of the standing waves and the resulting electron outbursts.

Unfortunately, this interpretation encountered a number of serious obstacles, such as a lack of the adequate boundary conditions in the freely-expanding plasma cloud without a sharp boundary. To get around this problem in paper[11], the outer boundary condition for the plasma waves was imposed at the radius where the electron Debye length was about the entire size of the plasma cloud. Although such an assumption might be reasonable in some approximation, it is rather crude. Really, the calculated TD resonances were found to depend appreciably on the above-mentioned position of the 'artificial wall'. Besides, it was unclear how the high-order TD harmonics—possessing a nontrivial angular dependence—could be excited by an almost uniform external electromagnetic field.

Moreover, the pattern of resonances was found to be weakly dependent on the initial electron energy and, therefore, the subsequent electron temperature. This fact was also rather suspicious from the viewpoint of TD paradigm, because the Bohm–Gross dispersion relation—resulting in the formation of TD resonances—crucially depends on the electron temperature. Finally, TD calculations should depend on the shape of the plasma cloud, while the experimental measurements[11] showed that the subharmonics of the electron emission remained almost the same even in the strongly distorted clouds, namely, when an opaque wire was placed in the ionizing laser beam, thereby resulting in the dumbbell-shaped initial distribution of the plasma density. (This test was originally em-

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ployed by the experimentalists to exclude the ion acoustic waves; but it is actually applicable to any type of the standing waves.)

Some of the above-listed problems were resolved in the course of the subsequent studies. In particular, as follows from the detailed experiment[14], even if a strip of the strongly-depleted density was originally formed in the plasma cloud, the charged-particle distribution heals over a few microsecond time scale, and the cloud restores its approximately Gaussian shape. So, it might be not so surprising that the electron resonances were almost insensitive even to the strong initial perturbations.

A considerable theoretical advance in the description of TD resonances was done in papers[15,16]. In particular, employing a dielectric permittivity of the collisional non-neutral plasmas, a remarkable similarity was found between the theoretical and experimental resonances; *e.g.*, Fig. 6 in the last-cited paper[16]. Unfortunately, these resonances were actually calculated for the heat absorbed by the plasma rather than for the escaping electrons. (In fact, the entire electron motion was assumed to be confined.) In general, it might be conjectured that the absorbed heat should stimulate an escape of electrons from the plasma cloud; but this effect does not follow immediately from the calculations. Besides, the pattern of resonances remained substantially dependent on the artificially-introduced cutoff radius of the cloud (the outer boundary condition); the problem being encountered already in the earlier study[11].

In view of the above circumstances, it seems reasonable to seek for the alternative explanations, which should immediately explain the release of electrons and be independent on the global properties of the plasma cloud. It is the aim of the present work to suggest one possible option. It is based on the radiowave ionization of the 'secondary' Rydberg atoms formed due to recombination in the expanding and cooling plasma cloud. Since this approach is purely 'local', it completely avoids any issues associated with boundary conditions and shape of the cloud. Besides, as will be seen from the subsequent simulations, this mechanism provides the patterns of peaks of the electron emission that are qualitatively similar to the observed ones.

II. THEORETICAL MODEL

A. Ionization–Recombination Balance in the Expanding Plasma Cloud

As was already mentioned above, we shall consider the ultracold plasma cloud expanding inside the rf resonator, as illustrated in Fig. 1. In the course of the expansion and cooling, the plasma decays into the Rydberg states due to three-body recombination, and the resulting Rydberg atoms are subsequently ionized again by the external electromagnetic field.

Assuming for simplicity that the cloud is uniform, the total flux of escaping electrons can be estimated as

$$I \propto \eta \left(dN_{\rm Ry}/dt\right)_{\rm rec} R^3, \tag{1}$$

where $N_{\rm Ry}$ is the concentration of the Rydberg atoms, η is the efficiency of ionization (*i.e.*, percentage of the ionized atoms), and R is the characteristic radius of the plasma cloud (for the additional discussion, see Appendix A).

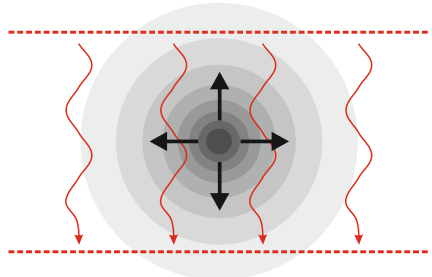

FIG. 1. Sketch of the plasma cloud expanding inside the rf resonator.

Next, the rate of collisional three-body recombination is given by the equation[17]:

$$\left(\frac{dN_{\rm Ry}}{dt}\right)_{\rm rec} \propto N_e^3\, T_e^{-9/2}, \tag{2}$$

where N_e is the concentration of the charged particles (*i.e.*, electrons and ions), and T_e is the electron temperature (which can be, in general, very different from the ionic one). Substituting formula (2) into (1), we get the total electron current in the form:

$$I \propto \eta\, N_e^3\, T_e^{-9/2}\, R^3. \tag{3}$$

At the inertial stage of expansion (which actually covers the most part of evolution of the plasma cloud), $R(t) \propto t$; so that

$$N_e \propto 1/R^3 \propto 1/t^3. \tag{4}$$

On the other hand, if the processes of inelastic scattering are assumed to be insignificant[1], then the electron temperature will decay by the adiabatic law with the adiabatic index $\gamma = 5/3$:

$$T_e \propto 1/R^{3(\gamma-1)} \propto 1/R^2 \propto 1/t^2. \tag{5}$$

Let us note that all quantities apart from η in formula (3) change monotonically with time. However, as will be shown in the next section, the efficiency of ionization η exhibits a strongly nonmonotonic behavior. Just this fact results in the observed multiple peaks of the total electron flux from the plasma cloud.

B. Efficiency of Ionization of Rydberg Atoms

The most pronounced series of peaks was observed under irradiation of the cloud by the electromagnetic waves with

[1] In fact, the evolution of electron temperature in ultracold plasmas can deviate substantially from the adiabatic law; *e.g.*, experiment[18] and its theoretical interpretation[19]. However, it is hardly reasonable to include such effects into the present simplified model.

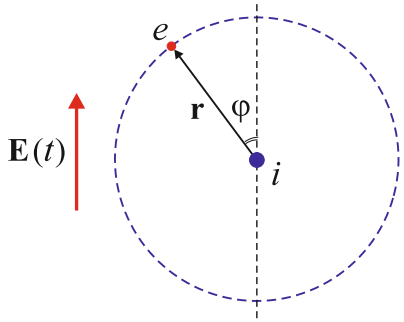

FIG. 2. Motion of the Rydberg electron (e) about the central ion (i) under the influence of the external electric field $\mathbf{E}$.

frequency 20–30 MHz, which corresponds to the photon energy 10^{-7} eV; while the characteristic binding energy of the Rydberg atoms with principal quantum numbers about 50—which are typically formed after recombination of ultracold plasmas—should be much greater, about 5×10^{-3} eV. Consequently, the ionization proceeds as a successive absorption of a very large number of photons and can be well described by the equations of classical mechanics and electrodynamics. We shall call it the quasi-classical multiphoton ionization. (Yet another term widely used in the literature—stochastic ionization—seems to be less appropriate and somewhat misleading, because such a process can often develop even without onset of any stochasticity.) Theoretical studies of this phenomenon were widely conducted starting from the late 1970's; *e.g.*, review[20]. The corresponding experiments were performed initially with the moderately-excited Rydberg atoms ($n \approx 10-30$) in the atomic beams irradiated by microwaves[21], so that such phenomenon was also called the microwave ionization.

The most important finding of the above-mentioned studies—both theoretical and experimental—was that the efficiency of ionization exhibits a number of well-expressed peaks, depending on the ratio of the applied irradiation frequency ω to the frequency of revolution of the Rydberg electron:

$$\omega/\Omega_n = \left(\hbar^3/m_e e^4\right) n^3\,\omega = \left(\omega/\Omega_1\right) n^3, \tag{6}$$

where $\hbar$ is the Planck constant, e and m_e are the electron charge and mass, n is the principal quantum number, and Ω_n is the 'classical' (Bohr) frequency of revolution of the electron. The frequency of revolution in the n'th orbit is evidently given by the relation:

$$\Omega_n = \frac{m_e e^4}{\hbar^3}\,\frac{1}{n^3} = \frac{\Omega_1}{n^3}. \tag{7}$$

Example of the sharp dependence of the efficiency of ionization on the ratio (6) can be seen, *e.g.*, in Fig. 3 in one of the earliest papers[22].

At last, a well-known property of the three-body recombination[17] is that it proceeds to the states with binding energies about the thermal energy of electrons T_e. As a result, the corresponding Rydberg atoms are formed in the states with the principal quantum numbers n^*:

$$\left|E_{n^*}\right| = \left|E_1\right|/n^{*2} \propto (3/2)\,A\,k\,T_e, \tag{8}$$

where k is the Boltzmann constant, A is a numerical coefficient on the order of unity, and E_n is the energy of the n'th atomic level. In other words,

$$n^* = \left(\frac{2\,|E_1|}{3Ak}\right)^{1/2} T_e^{-1/2} = \left(\frac{2\,|E_1|}{3Ak\,T_{e0}}\right)^{1/2} \left(t/t_0\right), \tag{9}$$

i.e., the principal quantum number of the Rydberg atoms increases linearly with time. (For simplicity, we assume here that all electrons at the given temperature T_e recombine to the single state n^*.)

Finally, taking into account relations (4)–(6), formula (3) for the total electron current can be rewritten as

$$I \propto \eta\left(n^*(t)\right) t^3. \tag{10}$$

C. Simulation of the Ionization

Unfortunately, despite a lot of the earlier results on the efficiency of quasi-classical multiphoton ionization of Rydberg atoms, it is impossible to apply them immediately to the problem under consideration. The case is that such studies usually treated the ionization as an escape of the electron up to 'infinity'. On the other hand, in the case of plasmas, if the electron is shifted from the original ion by a distance about a half of the mean interparticle separation, it begins to experience microfields of other particles and becomes 'collective' (ionized). So, the ionization of a Rydberg atom in plasmas should be treated as an escape of its electron up to some threshold radius $R_{\rm th}$, *i.e.*, actually its excitation to a higher orbit. Just this criterion is the crucial difference of our simulations from the early ones. As will be seen below, the overall pattern of the ionization peaks in such a case can be substantially modified.

The threshold radius can be estimated by the following way. Because of the above-mentioned property of the three-body recombination, the recombined atom is formed with a typical binding energy about the thermal energy of electrons, *i.e.*,

$$\frac{e^2}{r_{\rm Ry}} \approx \langle K_e \rangle, \tag{11}$$

where $r_{\rm Ry}$ is the characteristic radius of the Rydberg atom, and $\langle K_e \rangle$ is the average kinetic energy of an electron. On the other hand, this kinetic energy can be expressed through the typical potential energy of the plasma particles as

$$\langle K_e \rangle \approx \frac{1}{\Gamma_e}\,\frac{e^2}{\langle r \rangle}, \tag{12}$$

where Γ_e is the Coulomb coupling parameter for electrons, and $\langle r \rangle$ is the mean separation between the particles of the same sign. At last, substituting formula (12) to (11), we get the following relation:

$$r_{\rm Ry} \approx \Gamma_e\,\langle r \rangle. \tag{13}$$

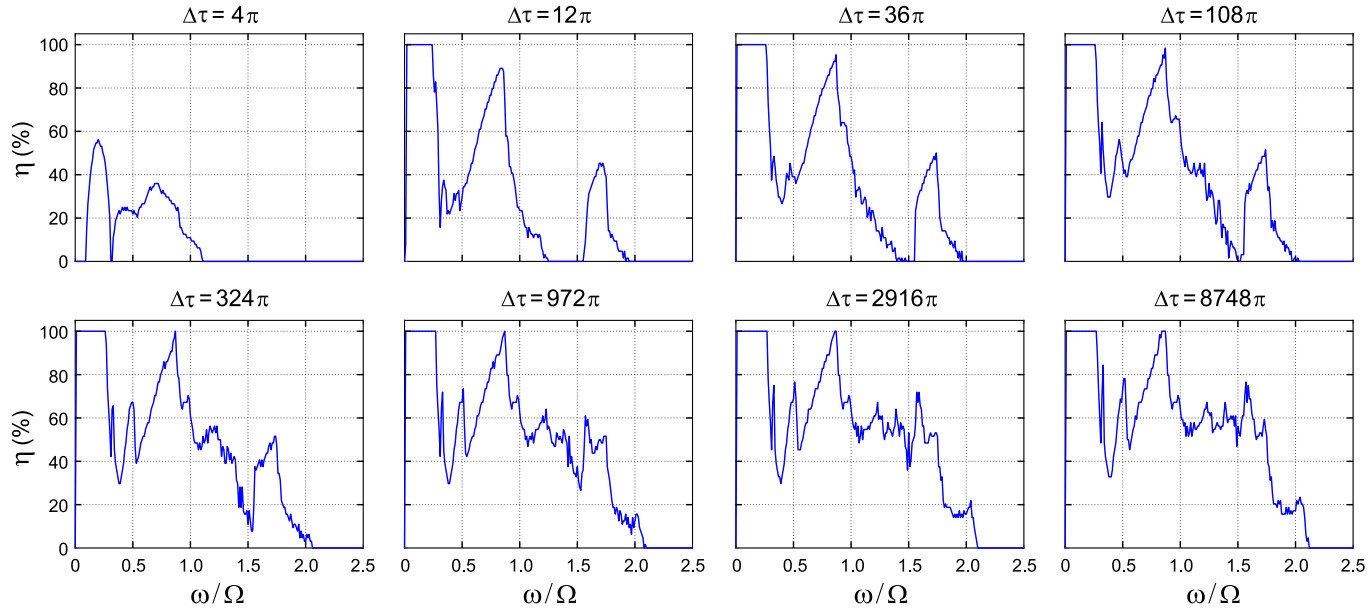

FIG. 3. Efficiency of ionization η as function of the ratio of the electromagnetic wave frequency ω to the frequency of revolution of the unperturbed Rydberg electron Ω. Different panels correspond to the different duration of interaction of the Rydberg atom with the electromagnetic wave $\Delta\tau$.

For example, if $\Gamma_e = 0.2$, then a typical size of the recombined Rydberg atom will be about one fifth of the characteristic interparticle separation. So, if an electron is located at the distance $2.5\,r_{\rm Ry}$ from the atomic center, it will experience approximately equal forces from the two adjacent ions. Therefore, the threshold radius $R_{\rm th}$ should be taken somewhat below this value. Particularly, in the most part of our calculations we used $R_{\rm th} = 1.5\,r_{\rm Ry}$. Anyway, as illustrated in Appendix C, the qualitative pattern of ionization (namely, positions of the ionization peaks) depends rather weakly on the exact value of $R_{\rm th}$.

Next, the equation of motion of the Rydberg electron perturbed by the external electric field $\mathbf{E}$ can be written as

$$m_e \frac{d^2\mathbf{r}}{dt^2} = -e^2\,\frac{\mathbf{r}}{r^3} - e\,\mathbf{E}(t), \tag{14}$$

where $\mathbf{r}$ is the electron radius vector, and the respective ion is assumed to be localized in the origin of coordinates, as illustrated in Fig. 2.

In the polar coordinate system, formula (14) is reduced to the set of equations:

$$m_e\big(\ddot{r} - r\,\dot{\varphi}^2\big) = -\frac{e^2}{r^2} - e\,E(t)\,\cos\varphi, \tag{15a}$$

$$m_e\big(r\,\ddot{\varphi} + 2\,\dot{r}\,\dot{\varphi}\big) = e\,E(t)\,\sin\varphi. \tag{15b}$$

Since the three-body recombination results in the formation of Rydberg atoms with large orbital quantum numbers l (about the principal quantum number n), we shall assume for simplicity that the initial unperturbed state was purely circular. So, the unperturbed solution of the above-written equations (*i.e.*, at $E \equiv 0$) will be

$$\varphi = \Omega t + \varphi_0, \qquad \Omega = \big(e^2/m_e\,a_0^3\big)^{1/2}, \tag{16}$$

where a_0 is the original unperturbed radius.

Next, let us introduce the dimensionless radius $\tilde{r}$ (marked by a tilde) and time τ:

$$r = a_0\,\tilde{r}, \qquad t = \Omega^{-1}\,\tau, \tag{17}$$

so that $\tau = 2\pi$ is the period of unperturbed revolution. Substitution of these definitions to the expressions (15a) and (15b) leads us to the dimensionless equations of motion:

$$m_e a_0 \Omega^2\big(\tilde{r}'' - \tilde{r}\,\varphi'^2\big) = -\frac{e^2}{a_0^2}\,\frac{1}{\tilde{r}^2} - e\,E(\tau)\,\cos\varphi\,, \tag{18a}$$

$$m_e a_0 \Omega^2\big(\tilde{r}\,\varphi'' + 2\,\tilde{r}'\,\varphi'\big) = e\,E(\tau)\,\sin\varphi\,, \tag{18b}$$

where prime denotes a differentiation with respect to τ.

Let the electromagnetic wave be monochromatic, with frequency ω and the initial phase ψ_0:

$$E(t) = E_0\sin\big(\omega t + \psi_0\big) = E_0\sin\Big(\frac{\omega}{\Omega}\tau + \psi_0\Big)\,. \tag{19}$$

Then, the set of equations (18a) and (18b) is reduced to

$$\tilde{r}'' - \tilde{r}\,\varphi'^2 = -\tilde{r}^{-2} - \tilde{E}_0\sin\big[(\omega/\Omega)\tau + \psi^*\big]\cos\varphi\,, \tag{20a}$$

$$\tilde{r}\,\varphi'' + 2\,\tilde{r}'\,\varphi' = \tilde{E}_0\sin\big[(\omega/\Omega)\tau + \psi^*\big]\sin\varphi\,, \tag{20b}$$

where the dimensionless amplitude of the electromagnetic wave is defined as

$$\tilde{E}_0 = \frac{eE_0}{m_e a_0 \Omega^2}\,. \tag{21}$$

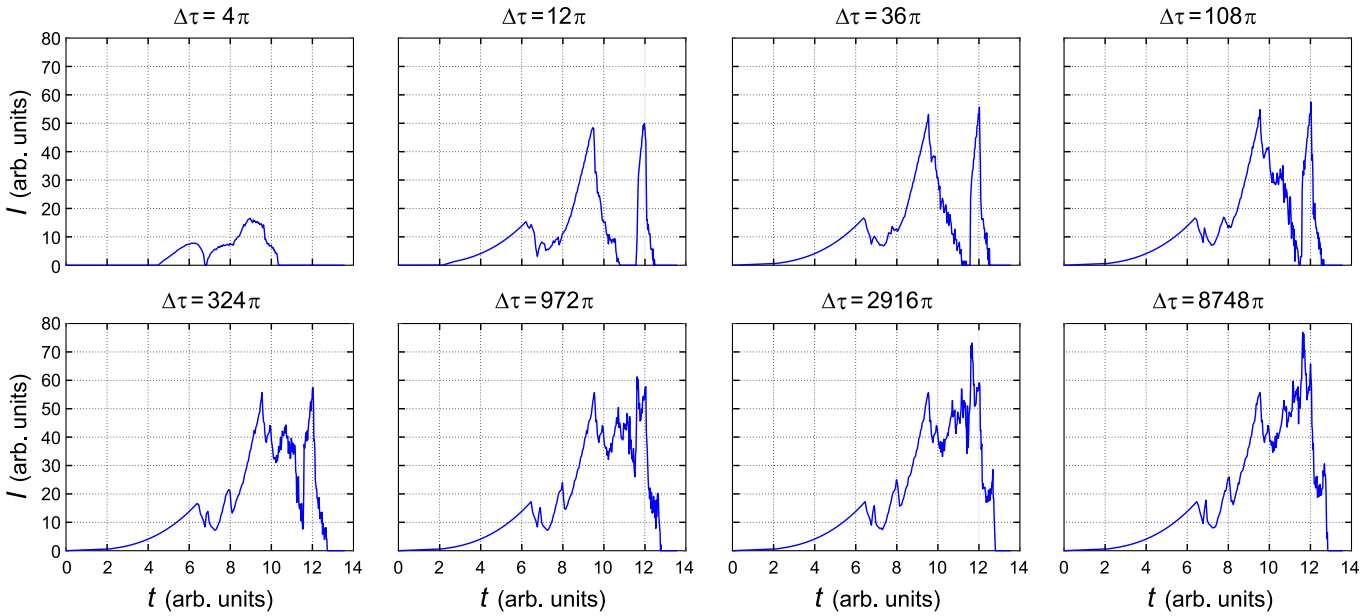

FIG. 4. Temporal dependences of the total electron current from the expanding plasma cloud at various durations of interaction of the Rydberg atoms with the electromagnetic wave.

To find the efficiency of ionization η, the set of equations (20a) and (20b) was solved numerically for a sufficiently large sample of the initial phases (namely, 64 uniformly distributed values). Then, η was calculated as a percentage of the initial phases when the computed orbit can reach the threshold radius $R_{\rm th}$ (*e.g.*, $1.5\,a_0$). As was already explained in the beginning of this section, when we deal with the ionization in plasmas, it is more reasonable to consider escape of an electron up to the limited distance rather than to infinity.

Results of the corresponding simulations are plotted in Fig. 3. They are presented for various duration of interaction between the Rydberg atom and the electromagnetic wave $\Delta\tau$. The amplitude of the external electromagnetic wave was taken to be $\tilde{E}_0 = 0.05$, *i.e.*, 5% of the characteristic interatomic field in the Rydberg atom. In fact, the subharmonics of electron emission are efficiently formed within some 'optimal' range of the field amplitude $\tilde{E}_0 \approx 0.02-0.10$; and we took the value approximately in the middle of this interval. (A more detailed discussion of the respective issue can be found in Appendix B.)

For a sufficiently short duration of interaction (*e.g.*, $\Delta\tau = 4\pi$), the plot in Fig. 3 is rather similar to the early studies[20,22], where the detached electron escaped to infinity. However, at the longer durations $\Delta\tau$ one can see the noticeable differences, namely, the ionization peaks in our simulations become more diverse and numerous.

At last, substituting the simulated dependences $\eta(\omega/\Omega)$ into formula (10) and taking into account (9), one can get the total electric current I as function of time in the course of the plasma cloud expansion. The corresponding plots for various $\Delta\tau$ are shown in Fig. 4. As is seen, a typical pattern of resonances is established starting from approximately $\Delta\tau = 12\pi$ (which corresponds to $\sim$6 wave periods), and then it changes rather gradually when $\Delta\tau$ increases further.[2]

Unfortunately, the characteristic duration of interaction between the Rydberg atom and the electromagnetic wave cannot be specified exactly in the framework of our simplified model, where the plasma cloud was assumed to be uniform, all Rydberg atoms formed in the same state (9), and the process of rf ionization occurred almost instantly (1). However, we can perform some estimates based on the remark in paper[11] that the patterns of resonances were almost unchanged when a continuous wave was replaced by the pulses of $1\,\mu$s duration. In other words, such a duration was sufficient for the rf ionization to complete.

For example, if we take the characteristic frequency of the external field $f = 10\,$MHz (starting from which one can observe multiple subharmonics in the experiment; see Fig. 1 in paper[11]), then the number of wave periods in the pulse of duration $\Delta t = 1\,\mu s$ will be $N_{\rm per} = \Delta t f \approx 10$. This value is in reasonable agreement with the remark in the previous paragraph that our simulations demonstrate a well-established pattern of subharmonics starting approximately from 6 wave periods. Next, if the experimental wave frequency f becomes greater, then the number of periods $N_{\rm per}$ within the same pulse duration Δt will increase. As is seen in the experimental plots, the resonances in such conditions become more numerous but closely located and 'shallow'. On the other hand, the simulations in Fig. 4 show that, when the number of periods in the pulse $N_{\rm per} = \Delta\tau/(2\pi)$ increases, the subpeaks of the total

[2] Strictly speaking, the dimensionless unit of time was defined in our simulations in terms of the inverse frequency of the unperturbed Rydberg electron Ω (17). However, since all subharmonics are formed at $\omega/\Omega \sim 1$ (within a factor of 2–2.5), we can assume for the crude estimates that the unit of time is associated with the inverse wave frequency ω.

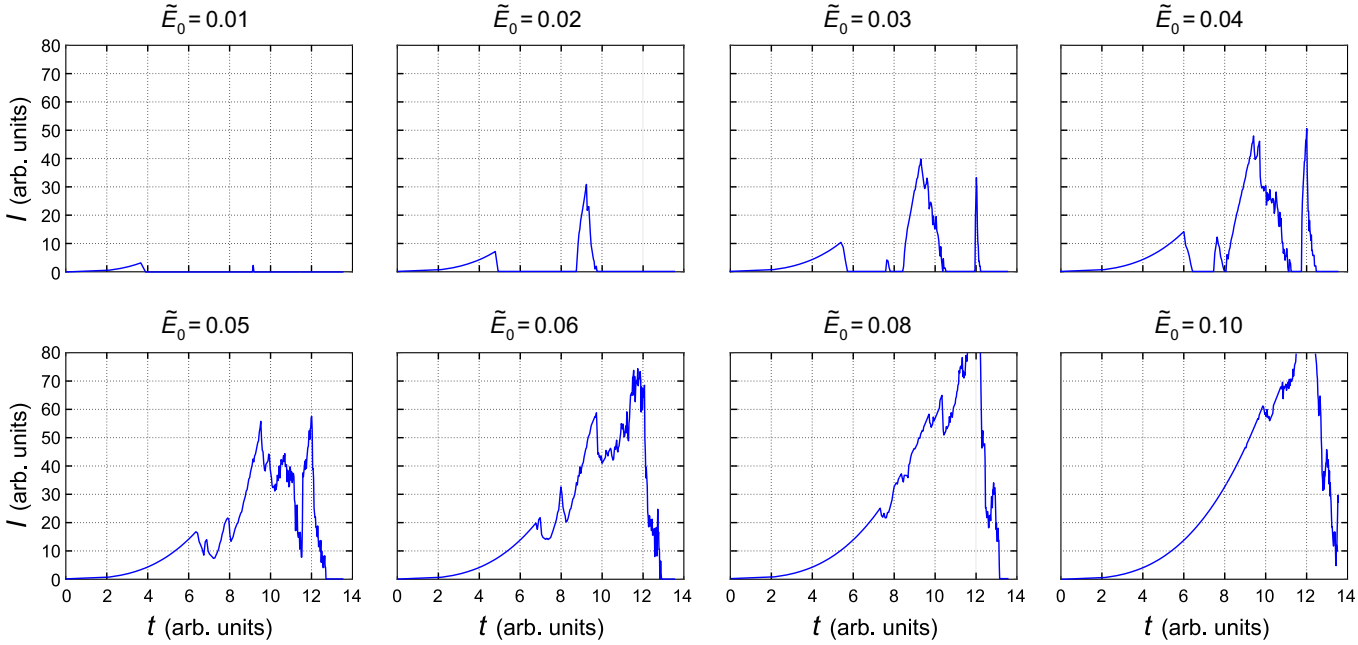

FIG. 5. Temporal dependences of the total electron current from the expanding plasma cloud at various amplitudes of the external electromagnetic wave $\tilde{E}_0$ (in the dimensionless units, normalized to the characteristic interatomic field (21)).

current also become more numerous but less distinctive (*i.e.*, begin to merge to each other).

Therefore, although our simplified model is unable to predict positions of the resonances as accurately as paper[16], the qualitative behavior of their amplitudes resembles the experimental patterns very well.

III. CONCLUSIONS

In summary, we proposed a new model of formation of subharmonics of the electron emission from the expanding UCP cloud irradiated by monochromatic radiowaves. This model is based on the idea of the quasi-classical multiphoton ionization of the 'secondary' Rydberg atoms formed due to the three-body recombination. Our model seems to be a reasonable alternative to the concept of Tonks–Dattner resonances, because is does not depend on the specific boundary conditions and is insensitive to the shape of the cloud. Besides, it naturally explains why the subharmonics of electron emission become more numerous but less expressed with the increasing frequency of irradiation. All these features are in good agreement with the experimental findings.

Of course, a much more detailed *ab initio* simulation should be performed to draw a definitive conclusion on the viability of this model. Such simulation should take into account both nonuniformity (*e.g.*, a Gaussian density profile) of the cloud and a nontrivial distribution of the Rydberg atoms over the various quantum states (as it was done, for example, in paper[23]).

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

DATA AVAILABILITY

The data that support the findings of this study are available from the authors upon reasonable request.

Appendix A: Derivation of the Expression for the Electric Current

At the first sight, expression (1) for the electric current might look a bit unusual, and a more reasonable choice would be the more standard formula:

$$I \propto k N_{\mathrm{Ry}} R^3, \tag{A1}$$

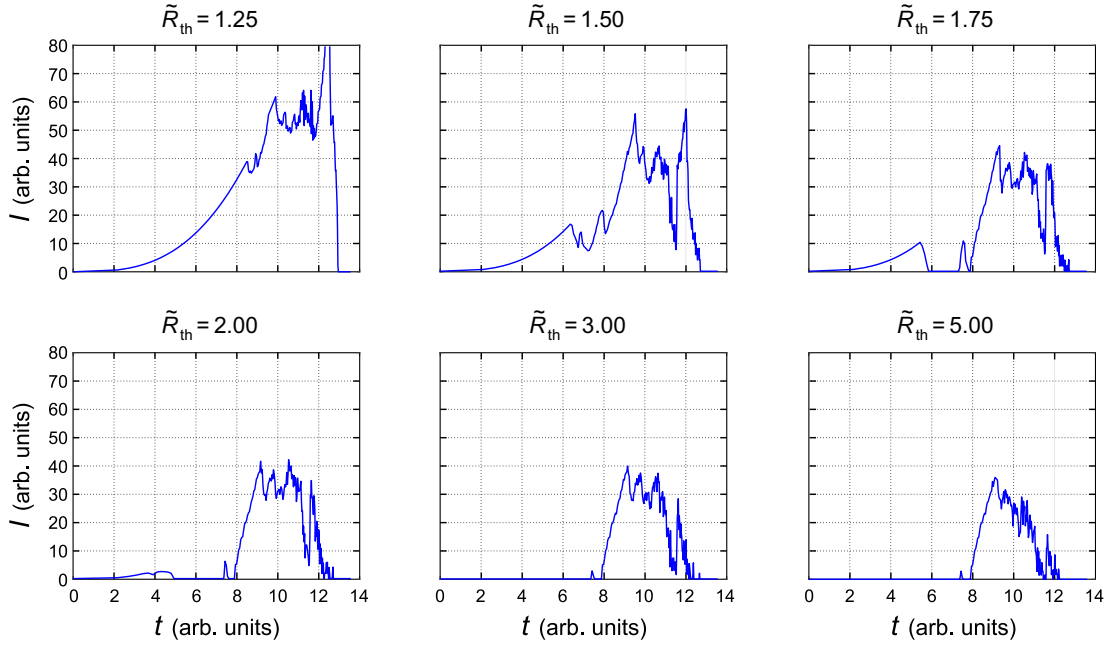

FIG. 6. Temporal dependences of the total electron current from the expanding plasma cloud for a set of the various threshold radii $\tilde{R}_{\rm th}$, at which the perturbed Rydberg atom is assumed to be ionized (in the dimensionless units, normalized to the size of the unperturbed atom a_0).

where $N_{\rm Ry}$ is the concentration of Rydberg atoms, R is the characteristic radius of the plasma cloud, and k is the probability of ionization of an atom in the unit of time.

However, it should be kept in mind that the above-written expression refers to the case of the ordinary single-photon ionization, when in each time interval a certain fraction of the available population of Rydberg atoms is ionized with probability k, leading to the exponential decay of that population in the course of time. But this is not the case of the multiphoton radiofrequency ionization, when the entire population is affected simultaneously by the electromagnetic wave. As a result, some fraction of these atoms (which is denoted by η) will be quickly ionized within a characteristic time interval Δt, while the remaining fraction $(1-\eta)$ collapse to the lower energetic states. So, the initial population will be quickly depleted. Meanwhile, a new population of the Rydberg atoms—which may have somewhat different physical properties (*e.g.*, the principal quantum number)—is formed at the same time interval due to the recombination.

The above-mentioned process can be described as follows. The number of free electrons released at the time interval Δt can be expressed, on the one hand, through the total electron current I and, on the other hand, through the available population of Rydberg atoms $\Delta N_{\rm Ry}(4\pi/3)R^3$ ionized with the efficiency η:

$$I\,\Delta t \propto \eta\,\Delta N_{\rm Ry} R^3. \tag{A2}$$

Next, dividing both sides of this relation by Δt and taking the formal limit $\Delta t \to 0$, we get exactly the equation (1).

Appendix B: Dependence of the Ionization Efficiency on the External Field Amplitude

In the main body of the present paper, the simulations were performed at the dimensionless amplitude of the electromagnetic wave $\tilde{E}_0 = 0.05$ (*i.e.*, 5% of the characteristic interatomic field), which was chosen somewhat arbitrary. So, it is interesting to check how sensitive are the corresponding results to the field amplitude.

The answer is given in Fig. 5, which represents simulations at a few different values of $\tilde{E}_0$, while the rf pulse duration is $\Delta\tau = 324\pi$, and all other parameters are the same as before. It is seen that separate peaks in the total electron current I become visible starting from $\tilde{E}_0 = 0.02$. Next, as the field amplitude increases, the peaks are more numerous but begin to merge with each other. As a result, we get the single major peak modulated by a few subpeaks. When the field amplitude increases further, this modulation becomes weaker, and finally it looks insignificant at $\tilde{E}_0 = 0.10$. In other words, the subharmonics of the electron emission are observable in a rather wide range of the rf field amplitudes, from 0.02 to 0.08; so that $\tilde{E}_0 = 0.05$ is a quite representative value.

Appendix C: Dependence of the Ionization Efficiency on the Threshold Radius

Yet another important issue in our simulations is the value of the threshold radius $\tilde{R}_{\rm th}$, starting from which the electron is assumed to experience appreciable microfields of other particles and, therefore, can be treated as quasi-free. In the main body of the paper, we used a rather small value $\tilde{R}_{\rm th} = 1.50$,

i.e., only 1.5 times greater than the radius of an unperturbed Rydberg atom. So, it is interesting to check how sensitive are our results to this parameter.

With this aim in view, we performed a series of additional simulations for various values of $\tilde{R}_{\rm th}$, while the parameters $\tilde{E}_0 = 0.05$ and $\Delta\tau = 324\pi$ were fixed. The corresponding results are presented in Fig. 6. It is seen that subharmonics of the total current I can be well identified already at $\tilde{R}_{\rm th} = 1.25$. Then, they become most pronounced in the range of the threshold radii $\tilde{R}_{\rm th} = 1.50-3.00$ (apart from the earliest-time peaks, which gradually disappear) and survive up to $\tilde{R}_{\rm th} = 5.00$. So, the value $\tilde{R}_{\rm th} = 1.50$—used in the most part of our simulations—gives a rater representative pattern of subharmonics.

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